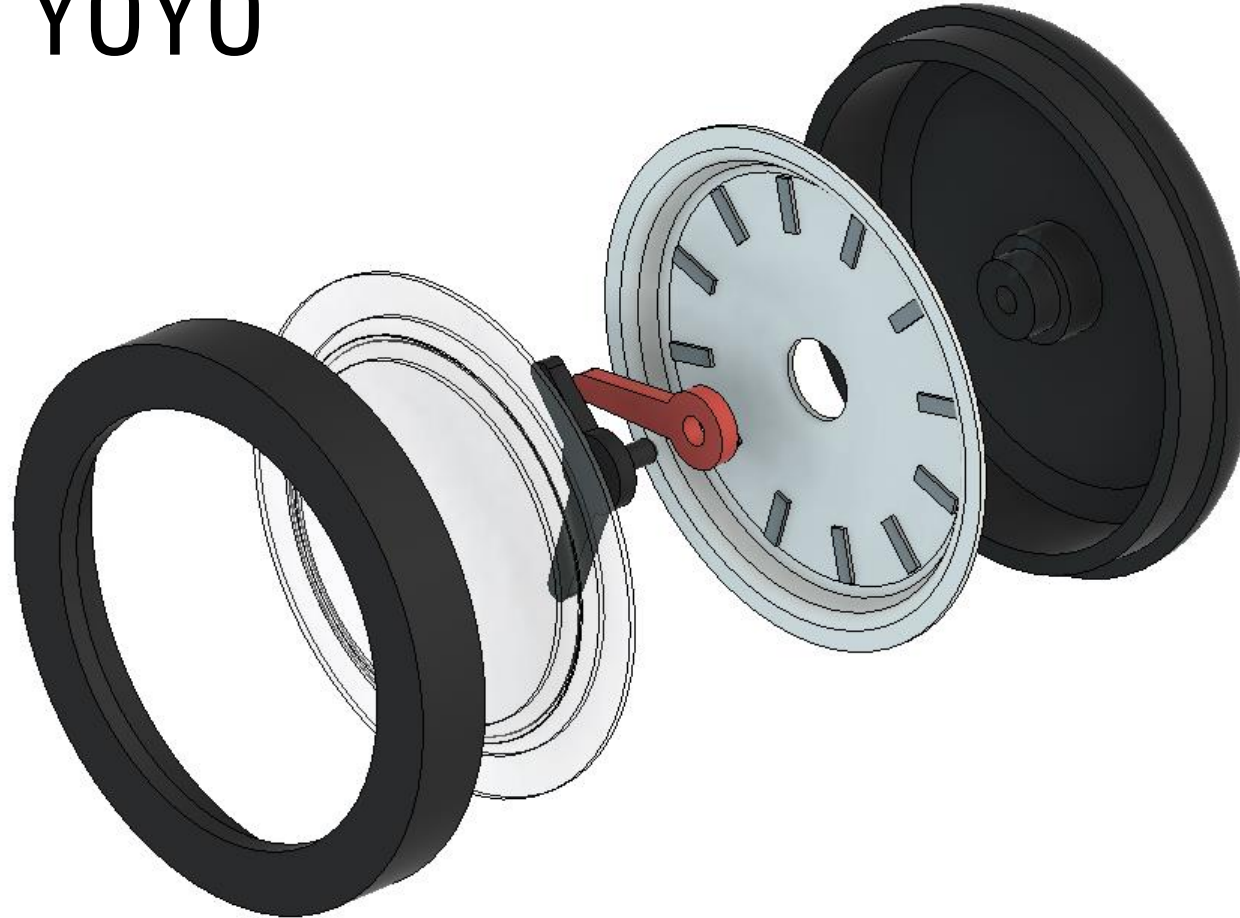

TEAM DEKA MANUFACTURING REVIEW

Alex Shanafield, Daniel Gliksberg, Emily Ha, Katherine Jankowski

THE CLOCK YOYO

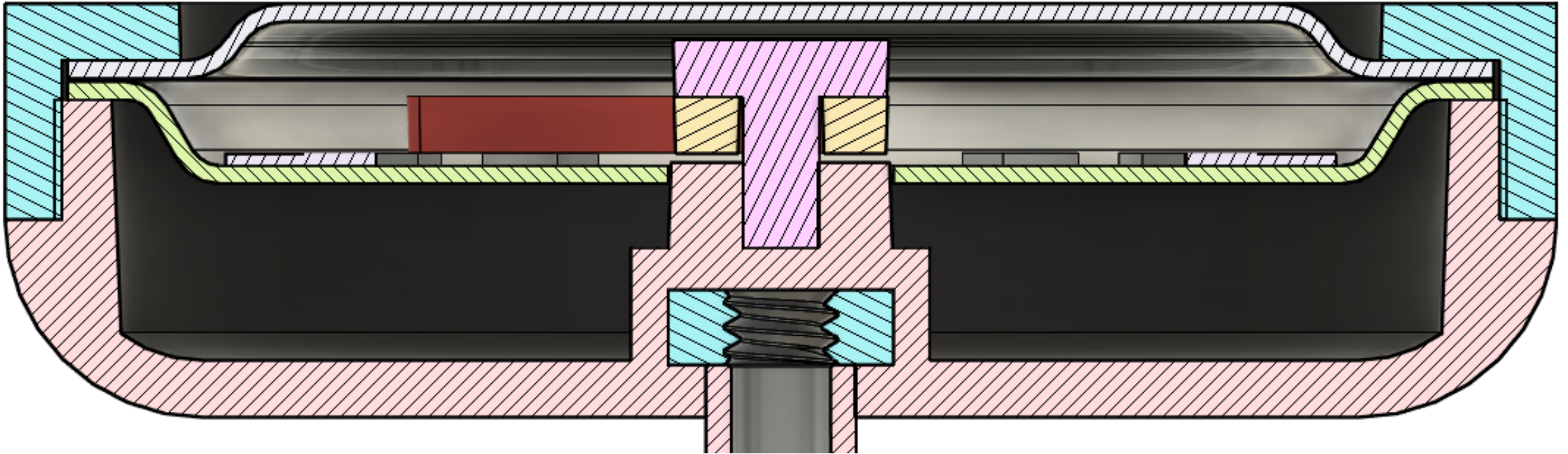


THE CLOCK YOYO



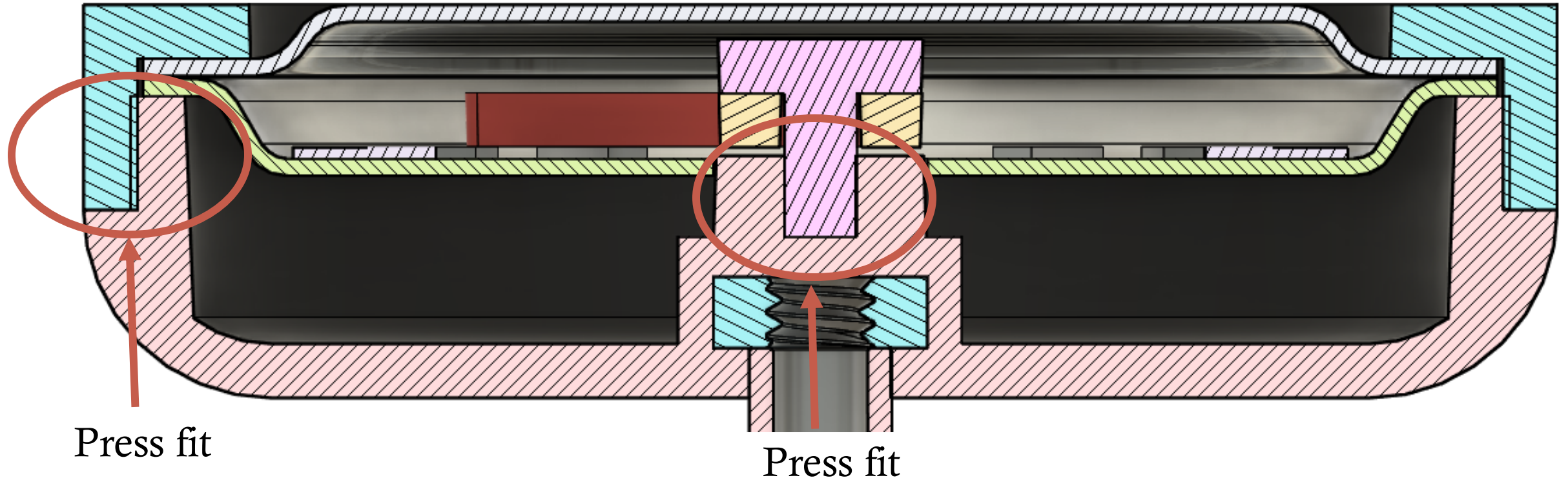
Week of	Objective	Risks
March 23rd	Finish machining all IM molds, all IM parts made at least once, all EF made at least once	All are first drafts, could be wrong geometry
March 30th	Update molds with learnings, have a fully assembled yoyo half	Could be a lot of learnings
April 5th	Add overmolded nuts	Could change how IM is working
April 13th	1 yoyo fully assembled	May need to tweak designs/specs
April 20th	Manufacturing Review, 25 yoyos fully assembled	Big ask/chunk of work to complete
April 27th	50 yoyos completed, our own drop test	Will have to remanufacture some if they fail
May 4th	All 50 yoyos assembled, pass the drop test!	

2 PRESS FITS



Note: For assembly images, parts are all shown un-scaled to show design intent.

2 PRESS FITS

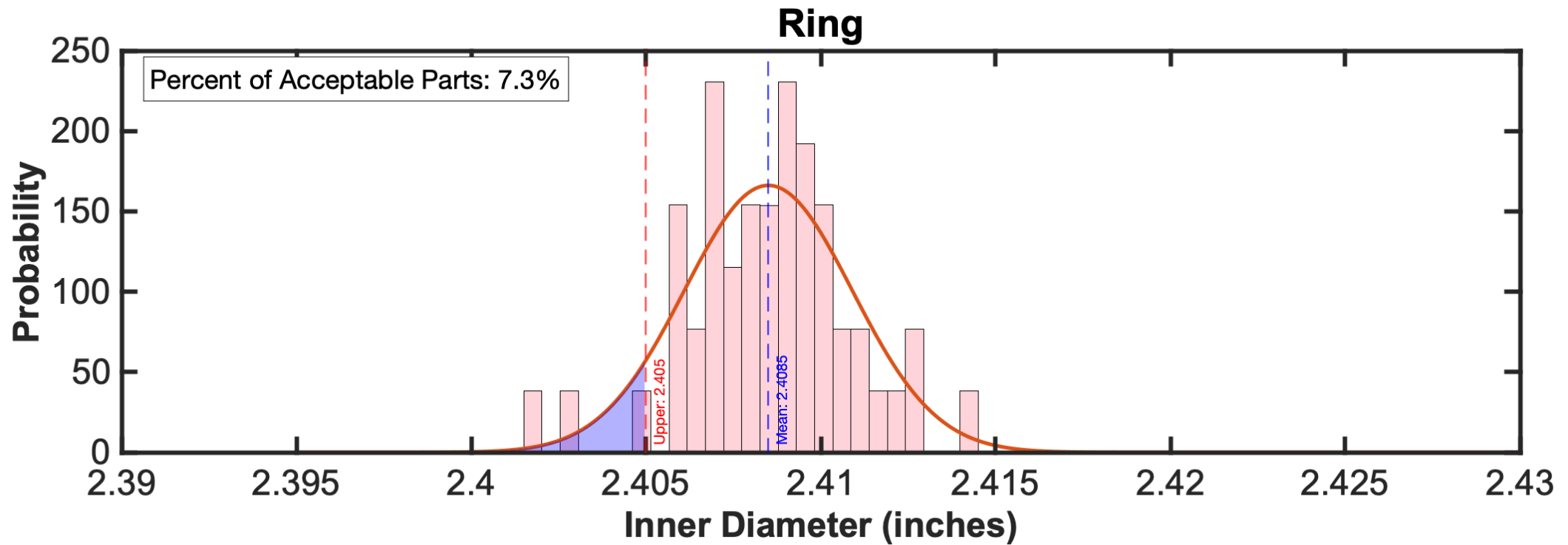


Note: For assembly images, parts are all shown un-scaled to show design intent.

RING MEASUREMENTS

Limits: [2.385, 2.405]

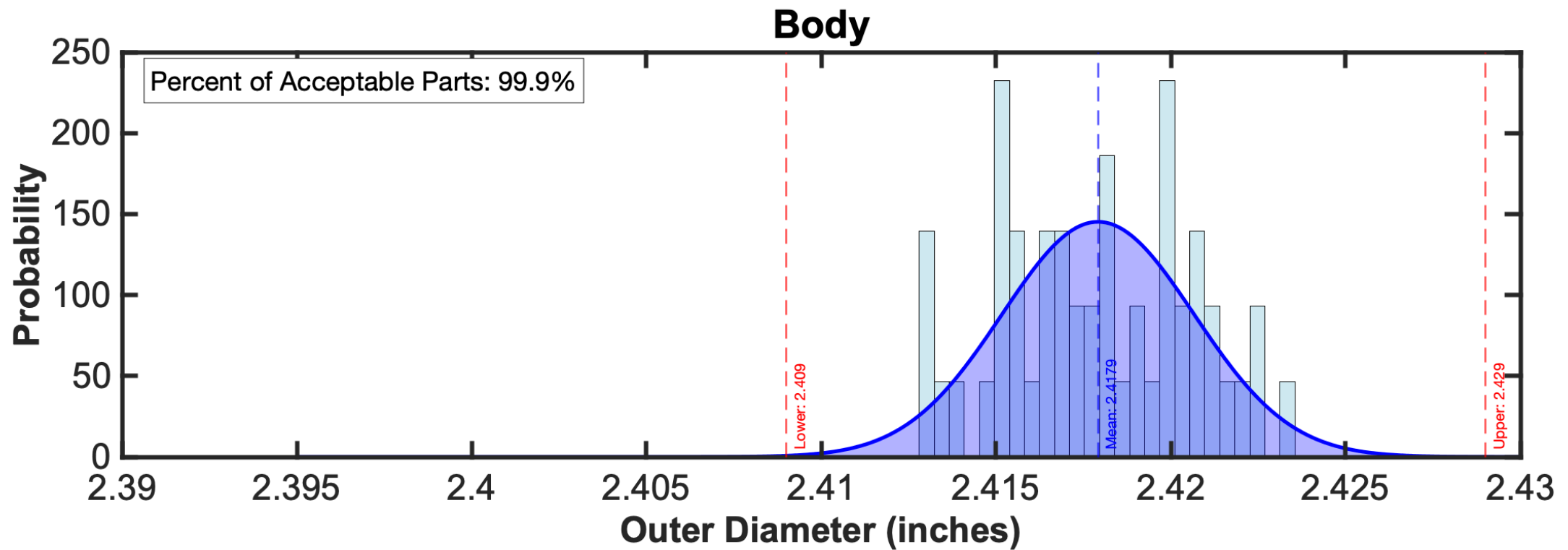
Mean: 2.4085 in



BODY MEASUREMENTS

Limits: [2.409, 2.429]

Mean: 2.4179 in

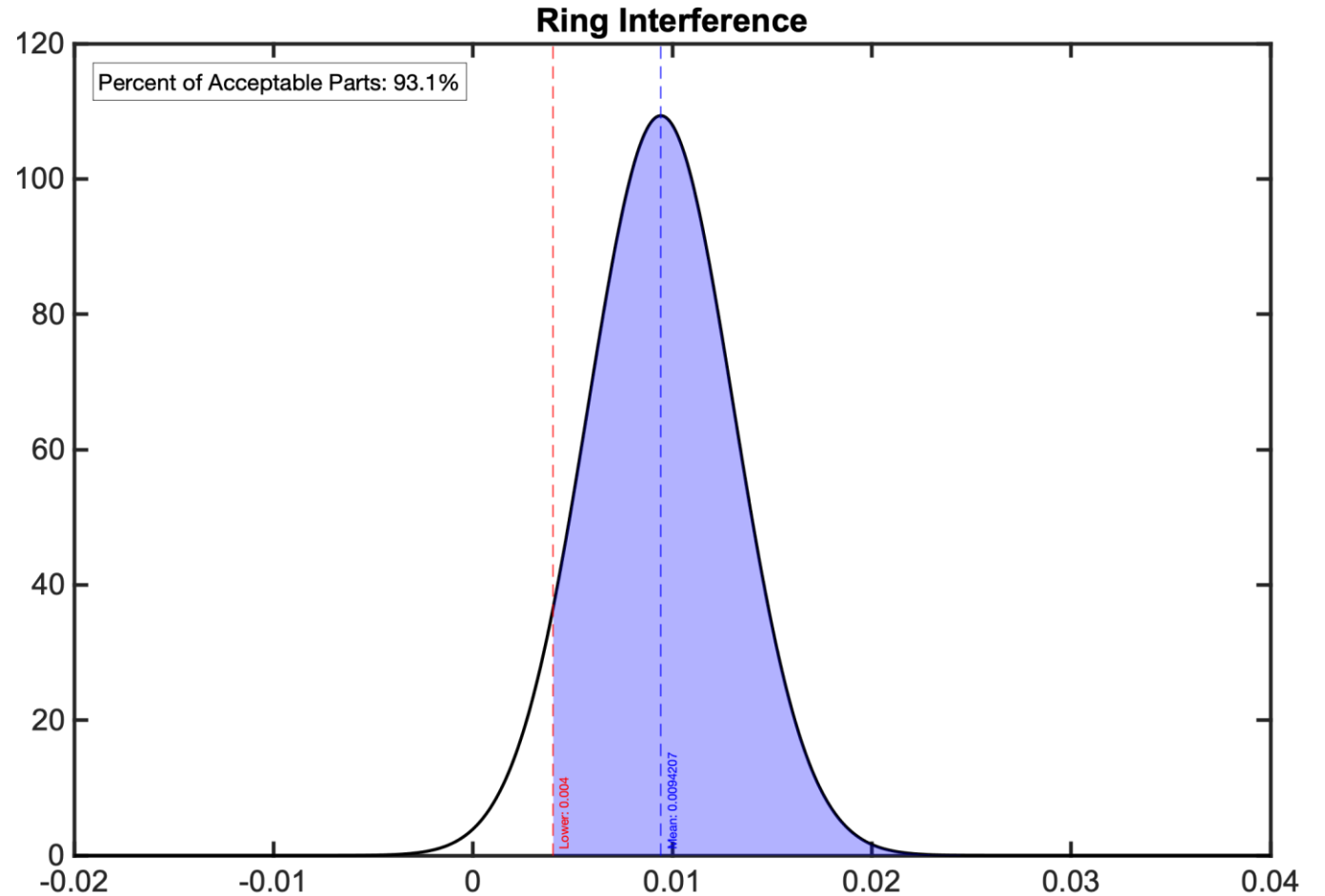


INTERFERENCE

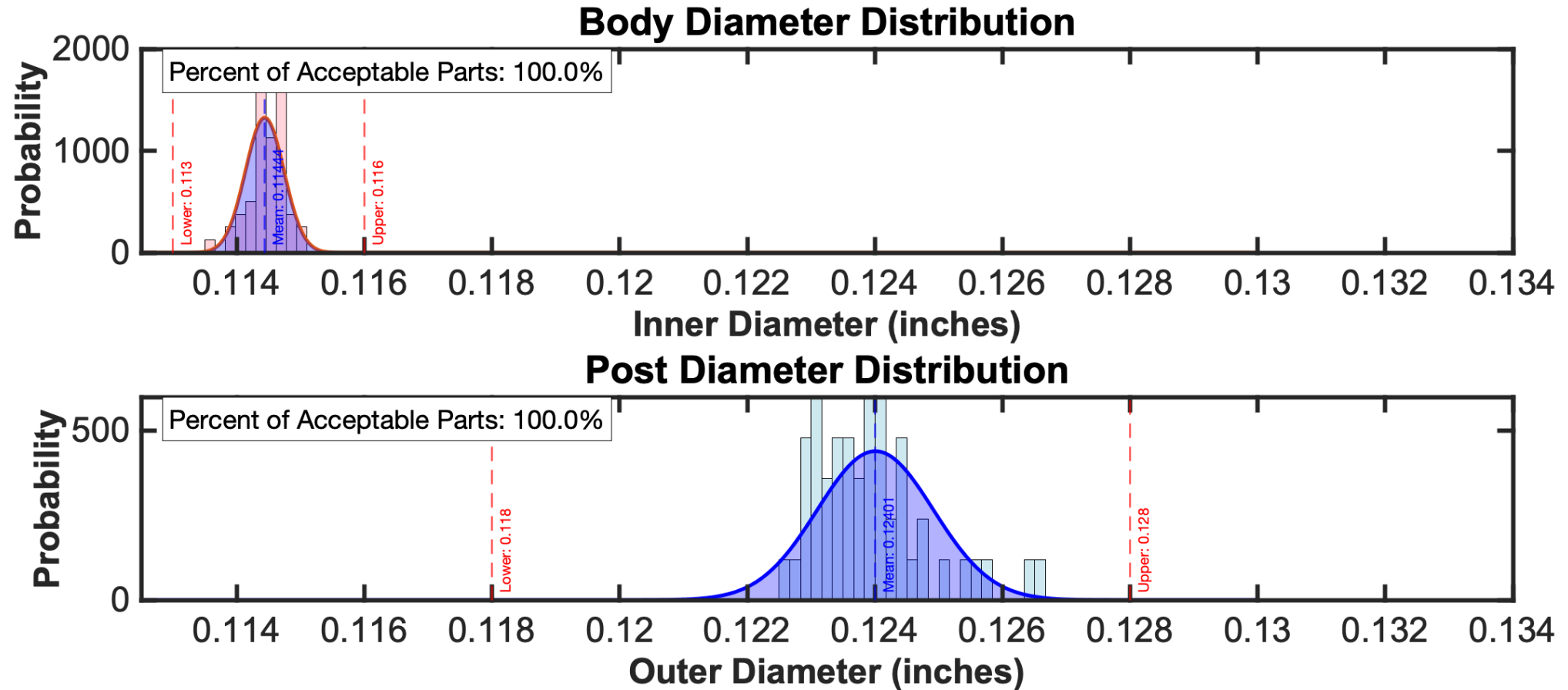
Limits: [0.006, 0.034]

Mean: 0.0094 in

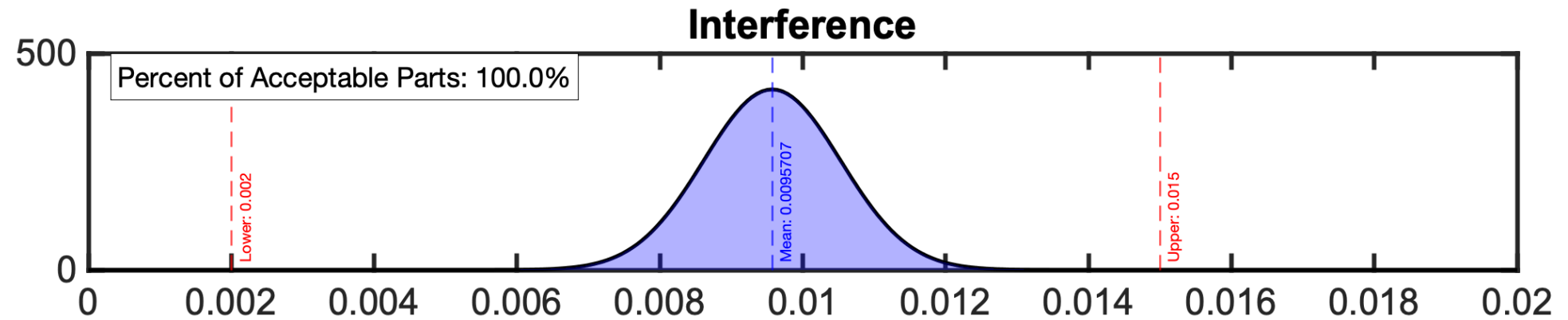
First Successful Press fit:



MINUTE HAND PRESS FIT



MINUTE HAND PRESS FIT



BODY MOLD MEASUREMENTS

Outer Diameter: 2.48046 in
Molded Body Mean: 2.4179 in

% Shrinkage: 2.52%

Inner Diameter: 0.11707 in
Molded Hole Mean: 0.11444 in

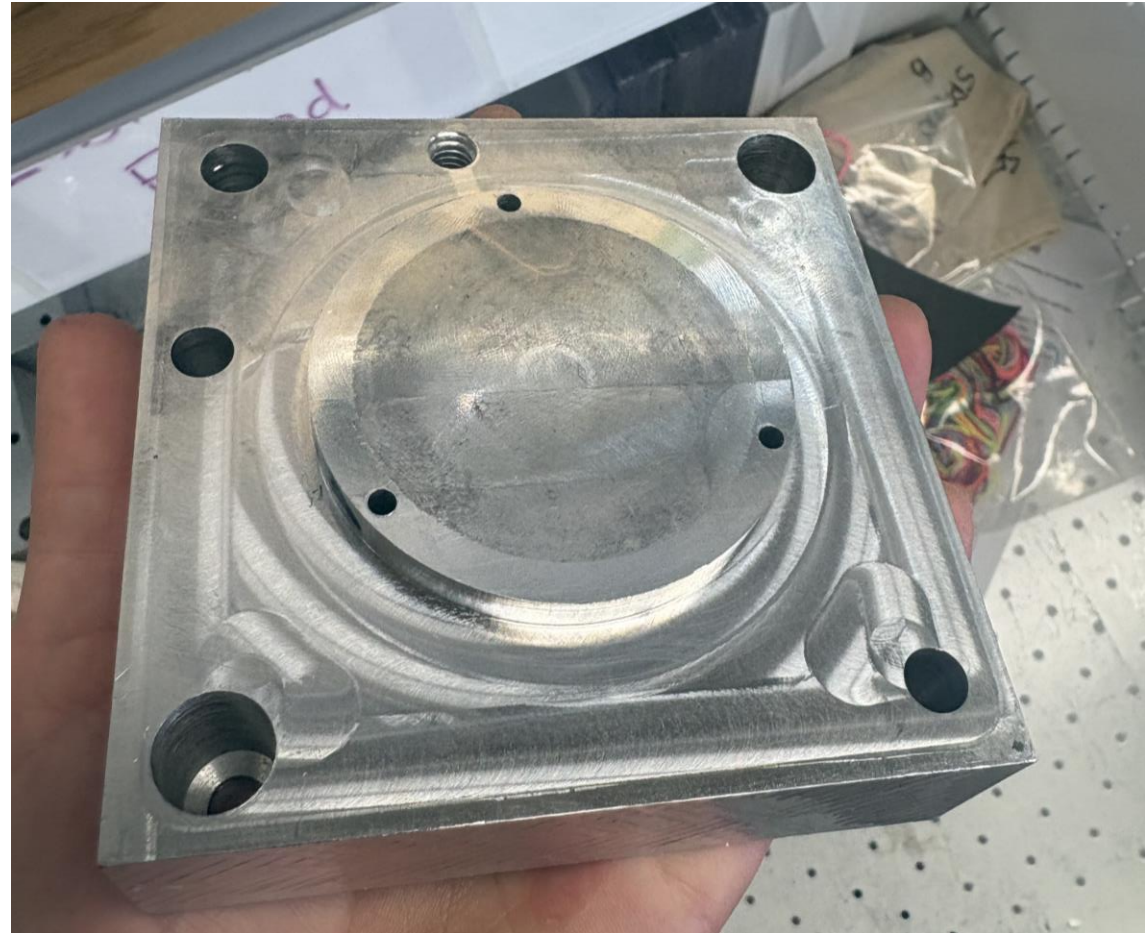
% Shrinkage: 2.24%



RING MOLD MEASUREMENTS

Inner Diameter: 2.43485 in
Molded Body Mean: 2.4085 in

% Shrinkage: 1.08%



CHALLENGES

MACHINING:

Ring mold:

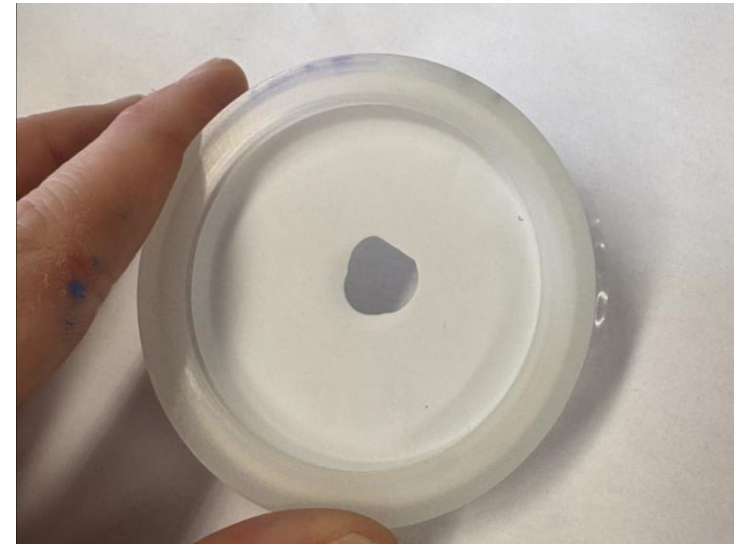
- Groove difficult to cut with long endmill (ended up breaking it 😞)
- Switched to shorter endmill
- Needed to machine several times to correctly account for shrinkage

Body mold:

- Needed to machine several times to achieve correct inner diameter for the minute hand pressfit
- Sanded down the surface to remove the CNC finish

INJECTION MOLDING:

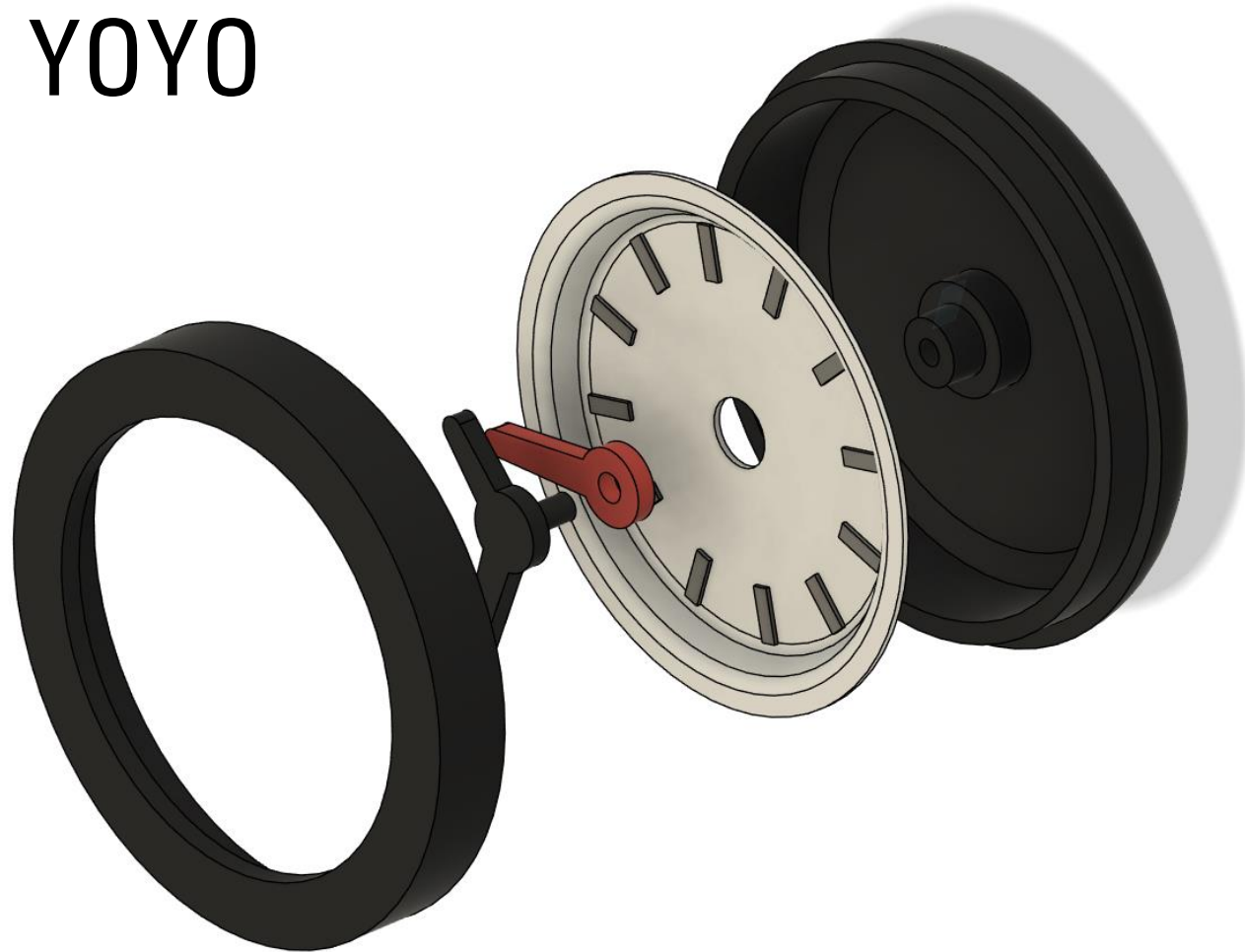
- Flash on Ring/Body:
 - Switched from BOY to Arbor (cut off any remaining flash with knife)
- Flash on minute hand post:
 - Post-process each with flush cutters



RISK AND MITIGATIONS

Part	Risk	Mitigation
Minute hand press fit	Press fit is not durable/doesn't work	Adjust TF geometry top bubble to hold it in, otherwise sticker
Two TFs between ring and body	Ring/body press fit is forced open	Adjust IM ring geometry or remove 1 TF part
Thermoformed clock face	Misalignment of the center hole	Make a jig to auto-align the part before punching a hole, or include a post in the mold that can later be more easily cut off to form a hole
	Running behind on schedule	Merge minute/hour hand with second hand as one part

THE CLOCK YOYO



STEPS TO 50 YOYOS – CURRENT POSITION

Part	Quantity Made
Body	104
Ring	104
Hour hand	109
Seconds hand	0
Bottom thermoform	2
Sticker	0

STEPS TO 50 YOYOS

Week of	Objective	Risks
April 20th	Bottom thermoformed plate x100, Seconds hand x100, start assembly	Big ask/chunk of work to complete
April 27th	Create/apply stickers, finish assembly, drop testing	Will have to remanufacture some if they fail
May 4th	All 50 yoyos assembled, pass the drop test!	

THANK YOU!
